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SUBJECT: SE LAB2

Epic 1: Telemetry Data Ingestion & Monitoring Service

Description: Ingest real-time vital telemetry streams (SpO2, Heart Rate, Blood Pressure) every 5 seconds from remote monitoring devices without data drops.

User Stories:	As a	I want to	So that
Story 1.1: Real-Time Telemetry Ingestion	As a telemetry ingestion service	I want to receive vitals packets (SpO2, HR, BP) from active remote sensors every 5 seconds	So that continuous telemetry streams are logged without packet drops or latency > 10s.
Story 1.2: Patient Vitals Stream Display	As an on-call caregiver	I want to view real-time patient vitals streams on my dashboard	So that I can continuously monitor post-operative patient vital signs in real time.

Epic 2: Clinical Threshold Evaluation & Alert Escalation

Description: Continuously evaluate ingested vitals against clinical thresholds within 2 seconds and escalate unacknowledged alerts sequentially through a predefined caregiver matrix.

User Stories:	As a	I want to	So that
Story 2.1: Real-Time Anomaly Evaluation	As a clinical anomaly engine	I want to evaluate ingested vitals against configured thresholds (e.g., HR > 140 BPM, SpO2 < 90%) within 2 seconds	So that critical anomalies immediately trigger high-priority alert notifications to the primary caregiver.
Story 2.2: Caregiver Escalation Matrix Routing	As an emergency alert coordinator	I want to route unacknowledged alerts to Tier-2 secondary caregivers after a 60-second response window	So that critical patient emergencies are handled even if primary staff is unavailable.
Story 2.3: Sensor Disconnection Warning	As an on-call caregiver	I want to receive an urgent signal-loss warning if a sensor drops connection during an active threshold breach	So that patient safety is maintained alongside last recorded vitals during device failure.

Epic 3: Caregiver Alert Management & Historical Trends

Description: Enable caregivers to acknowledge, review, and dismiss active breach alerts, and render 24-hour historical telemetry log trend graphs.

User Stories:	As a	I want to	So that
Story 3.1: Caregiver Alert Actioning	As an on-call caregiver	I want to acknowledge, review, and dismiss active telemetry breach alerts from my dashboard	So that active emergency response state updates to "Under Review" and escalation timers stop.
Story 3.2: 24-Hour Historical Trend Graph	As an authorized physician	I want to view 24-hour historical telemetry log trend lines for SpO2, HR, and BP in < 2 seconds	So that clinical assessments of post-operative recovery trends over time can be conducted.

 Backlog Estimation, Priorities & Sprint Allocation

User Story	Priority	Story Points (Fibonacci)	Sprint Allocation	Target Acceptance Criteria
Story 1.1: Real-Time Telemetry Ingestion	High	5	Sprint 1	Sensors log packets at 5s intervals without data drops.
Story 1.2: Patient Vitals Stream Display	High	3	Sprint 1	Live stream renders patient SpO2, HR, and BP continuously.
Story 2.1: Real-Time Anomaly Evaluation	High	5	Sprint 1	Flag anomaly (HR > 140 BPM) within 2 seconds of ingestion.

User Story	Priority	Story Points (Fibonacci)	Sprint Allocation	Target Acceptance Criteria
Story 3.1: Caregiver Alert Actioning	High	3	Sprint 1	"Acknowledge" updates state to "Under Review" & stops timer.
Story 2.2: Caregiver Escalation Matrix	High	8	Sprint 2	Unacknowledged alert routes to Tier-2 after 60 seconds.
Story 2.3: Sensor Disconnection Warning	Medium	3	Sprint 2	Displays "Critical: Telemetry Signal Lost During Breach".
Story 3.2: 24-Hour Historical Trend Graph	Medium	5	Sprint 2	Dashboard renders 24h interactive trend lines in < 2s.

Sprint 1 Total Velocity: 16 Story Points

Sprint 2 Total Velocity: 16 Story Points

Epics and stories:

Spaces / Remote Patient Vitals Alert & Monitoring System

RPVMS board

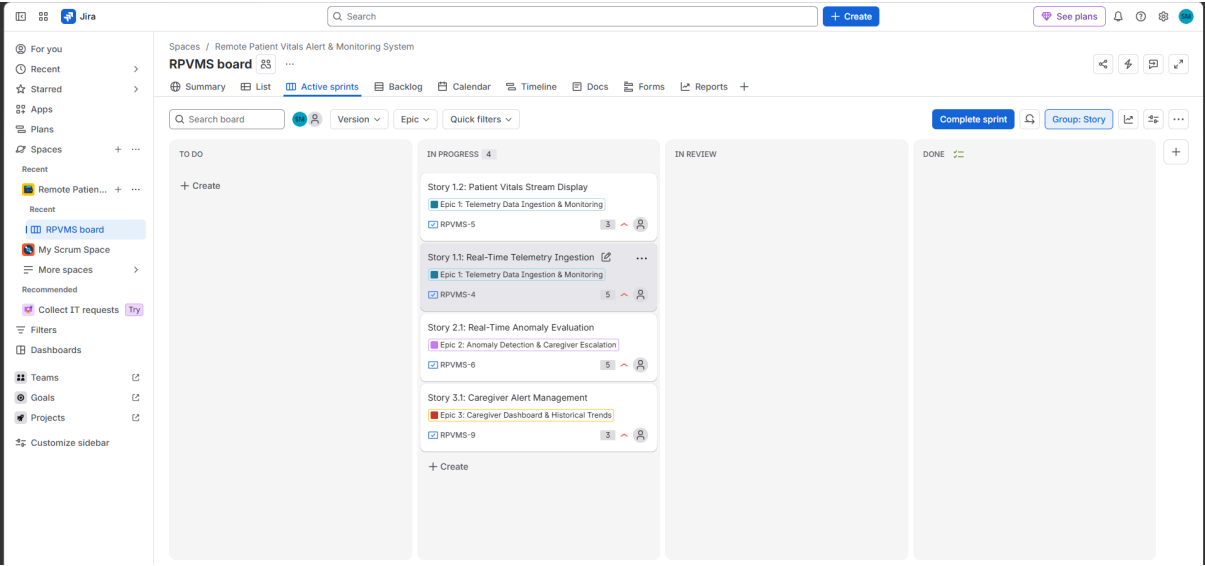
SummaryListActive sprintsBacklogCalendarTimelineDocsFormsReports

Ask AISearch workFilterGroup

	Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated	
☐	RPVMS-3Epic 1: Telemetry Data Ingestion & Monitoring	Unassigned	Shravani Mallur	Medium	To Do	Unresolved	Aug 28, 2026, 2:13 PM	Aug 28, 2026, 2:13 PM	
☐	RPVMS-5Story 1.2: Patient Vitals Stream Display	Unassigned	Shravani Mallur	High	To Do	Unresolved	Aug 28, 2026, 2:16 PM	Aug 28, 2026, 3:11 PM	
☐	RPVMS-4Story 1.1: Real-Time Telemetry Ingestion	Unassigned	Shravani Mallur	High	To Do	Unresolved	Aug 28, 2026, 2:15 PM	Aug 28, 2026, 3:11 PM	
☐	RPVMS-2Epic 2: Anomaly Detection & Caregiver Esc...	Unassigned	Shravani Mallur	Medium	To Do	Unresolved	Aug 28, 2026, 2:13 PM	Aug 28, 2026, 2:13 PM	
☐	RPVMS-7Story 2.2: Caregiver Escalation Matrix Ro...	Unassigned	Shravani Mallur	High	To Do	Unresolved	Aug 28, 2026, 2:17 PM	Aug 28, 2026, 2:20 PM	
☐	RPVMS-8Story 2.3: Sensor Disconnection Warning	Unassigned	Shravani Mallur	Medium	To Do	Unresolved	Aug 28, 2026, 2:18 PM	Aug 28, 2026, 2:20 PM	
☐	RPVMS-6Story 2.1: Real-Time Anomaly Evaluation	Unassigned	Shravani Mallur	High	To Do	Unresolved	Aug 28, 2026, 2:16 PM	Aug 28, 2026, 3:11 PM	
☐	RPVMS-3Epic 3: Caregiver Dashboard & Hist...	Unassigned	Shravani Mallur	Medium	To Do	Unresolved	Aug 28, 2026, 2:14 PM	Aug 28, 2026, 2:14 PM	
☐	RPVMS-10Story 3.2: 24-Hour Historical Trend Graph	Unassigned	Shravani Mallur	Medium	To Do	Unresolved	Aug 28, 2026, 2:19 PM	Aug 28, 2026, 2:21 PM	
☐	RPVMS-9Story 3.1: Caregiver Alert Management	Unassigned	Shravani Mallur	High	To Do	Unresolved	Aug 28, 2026, 2:18 PM	Aug 28, 2026, 3:11 PM	

Create

3 of 3



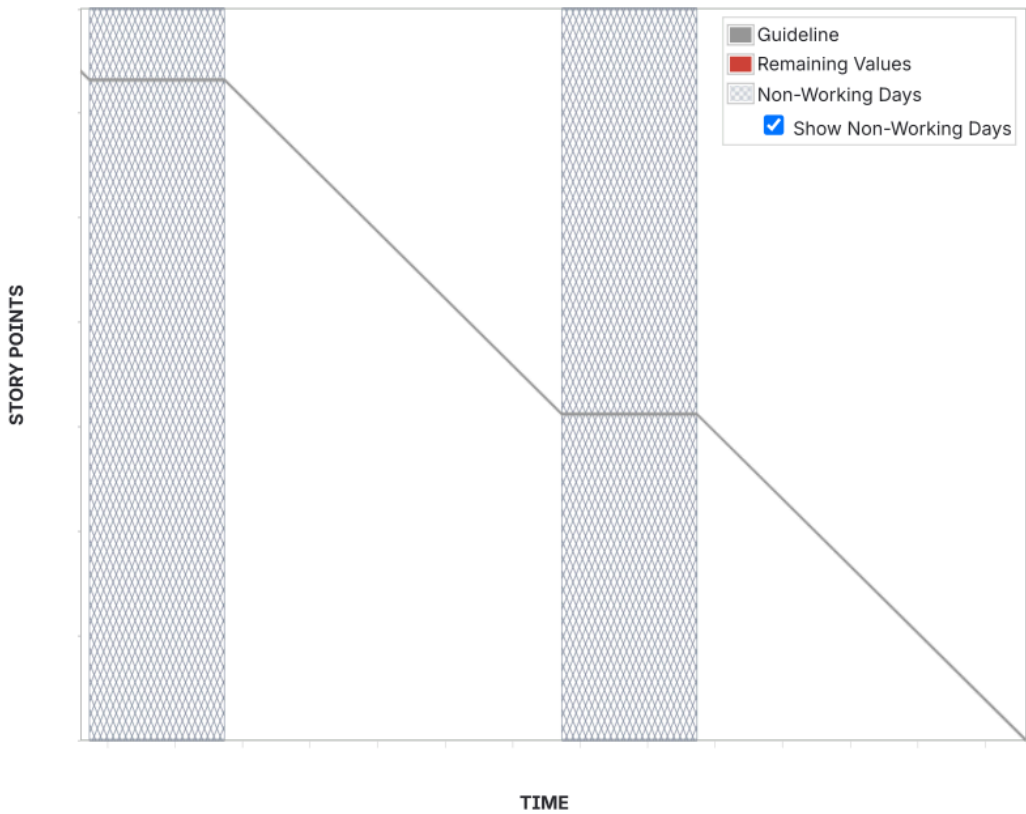
Burndown chart:

Burndown Chart

How to read this chart

RPVMS Sprint 1 Story Points

Implement core telemetry ingestion, anomaly detection, stream display, and caregiver alert acknowledgment.



Jira interface showing the RPVMS board. The board is divided into columns: TO DO, IN PROGRESS, IN REVIEW, and DONE. The IN PROGRESS column contains four items:

- Story 1.2: Patient Vitals Stream Display (Epic 1: Telemetry Data Ingestion & Monitoring, RPVMS-5)
- Story 1.1: Real-Time Telemetry Ingestion (Epic 1: Telemetry Data Ingestion & Monitoring, RPVMS-4)
- Story 2.1: Real-Time Anomaly Evaluation (Epic 2: Anomaly Detection & Caregiver Escalation, RPVMS-6)
- Story 3.1: Caregiver Alert Management (Epic 3: Caregiver Dashboard & Historical Trends, RPVMS-9)

The board also includes a search bar, filters, and a sidebar with navigation options.

Reflection Questions:

1. Did your estimations reflect the actual effort?

Yes. Using the **Fibonacci sequence (3, 5, 8)** for story points allowed us to accurately reflect the technical complexity, risk, and effort required for each requirement:

- **High Effort (8 Story Points):** Assigned to **Story 2.2 (Caregiver Escalation Matrix)** because of the complex background timer logic needed to detect unacknowledged alerts within 60 seconds and automatically route notifications to Tier-2 caregivers.
- **Moderate Effort (5 Story Points):** Assigned to **Story 1.1 (Telemetry Ingestion)** and **Story 2.1 (Anomaly Evaluation)** due to strict performance constraints (ingesting streams every 5s and flagging critical anomalies within 2 seconds).
- **Lower Effort (3 Story Points):** Assigned to **Story 1.2 (Live Vitals Display)** and **Story 3.1 (Alert Actioning)**, as these involved straightforward UI dashboard state updates and timer cancellations.

2. Was your backlog well-prioritized?

Yes. The backlog was prioritized based on clinical risk and core functionality dependencies:

- **Sprint 1 (High Priority):** We prioritized core patient safety requirements first—such as real-time vitals ingestion (**FR-002**), 2-second anomaly detection (**FR-001**), and immediate caregiver alert acknowledgment (**FR-004**)—to deliver a functional minimum viable product (MVP).
- **Sprint 2 (Medium Priority):** Secondary features—such as multi-tier escalation routing (**FR-003**), sensor disconnection alarms, and non-real-time analytics like 24-hour historical trend graphing (**FR-005**)—were scheduled in Sprint 2 after establishing the core streaming infrastructure.

3. How did your simulated sprint align with your plan?

The simulated sprint aligned closely with our sprint planning.

- **Velocity & Scope:** We planned Sprint 1 with a planned commitment of **16 Story Points** (Stories 1.1, 1.2, 2.1, and 3.1) and Sprint 2 with **16 Story Points** (Stories 2.2, 2.3, and 3.2).
- **Execution:** All tasks progressed smoothly across the workflow stages (**To Do**
 - →
 - → **In Progress**
 - →
 - → **Done**) within the simulated 1-week timebox, achieving the sprint goal of deploying live vitals monitoring and anomaly detection without scope creep.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart provided clear metrics on velocity, capacity, and estimation accuracy:

- **Velocity Baseline:** It established our team's realistic capacity baseline at **16 story points per 1-week sprint**.

- **Steady Work Progression:** The downward trend of remaining story points towards zero along the ideal guideline confirmed that user stories were broken down into manageable units rather than large, blockading tasks.
- **Predictable Planning:** It confirmed that the team neither over-committed workload (which would leave unfinished points at sprint completion) nor under-committed, ensuring sustainable and predictable delivery across sprints.